

Question

Write a menu-driven Java program to demonstrate basic programming constructs such as variables, data types, operators, conditional statements, loops, and user-defined methods by developing a *Student Utility Program*.

Steps for Execution

1. Open your IDE (Eclipse/VS Code) or Command Prompt.
2. Create a new Java file named `StudentUtilityProgram.java`.
3. Write the complete program code inside the file.
4. Save the file.
5. Compile the program using: `javac StudentUtilityProgram.java`
6. Run the program using: `java StudentUtilityProgram`
7. Enter inputs as prompted and test all menu options.

Algorithm

1. Start the program.
2. Display a menu with options for student info, conditionals, loops, and methods.
3. Read user choice.
4. Use switch to perform the selected operation.
5. **Part A:** Read student details, calculate total and percentage, display grade.
6. **Part B:** Perform even/odd check, largest of three, grade display, day of week.
7. **Part C:** Print multiplication table, numbers 1 to N, sum of natural numbers, Fibonacci series.
8. **Part D:** Call methods for factorial, prime check, max of two numbers, and area of circle.
9. Repeat menu until user chooses Exit.
10. Stop the program.

UML Class Diagram

```
+-----+
| StudentUtilityProgram |
+-----+
| + displayInfo()      |
| + calTotal()         |
| + calPercentage()    |
| + evenOdd()          |
| + findLargest()      |
| + displayResult()    |
| + dayOfWeek()        |
| + printTable()       |
| + displayNumbers()   |
| + sumNaturalNumbers() |
| + fibonacciSeries()  |
| + factorial()        |
| + isPrime()          |
| + maxOfTwo()         |
| + areaOfCircle()     |
+-----+
| + main()             |
```

CODE: SRC

```
package studentutility;

import java.util.Scanner;

public class StudentUtilityProgram {

    // Part A: basics
```

```

        public static void displayInfo(String name, int roll,int[] marks) {
            int total = calTotal(marks);
            double percentage = calPercentage(total, marks.length);
            System.out.println("\nStudent information : ");
            System.out.println("Name of the student: " + name);
            System.out.println("Roll number: " + roll);
            System.out.println("Total marks: " + total);
            System.out.println("Percentage: " + percentage );
            displayResult(percentage);
        }
    public static int calTotal(int[] marks) {
        int total = 0;
        for(int m : marks)
            total += m;
        return total;
    }
    public static double calPercentage( int total, int subjects) {
        return (double) total / subjects;
    }
}

// conditional statements

public static void evenOdd(int num) {
    System.out.println(num % 2 == 0 ? num + " is Even" : num + " is Odd");
}

public static void findLargest(int a, int b, int c) {
    int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
    System.out.println("Largest number is: " + largest);
}

```

```

    }

    public static void displayResult(double percentage) {
        if (percentage >= 90) System.out.println("Grade: A+");
        else if (percentage >= 75) System.out.println("Grade: A");
        else if (percentage >= 60) System.out.println("Grade: B");
        else if (percentage >= 40) System.out.println("Grade: C");
        else System.out.println("Grade: Fail");
    }

```

```

    public static void dayOfWeek(int dayNumber) {
        switch (dayNumber) {
            case 1: System.out.println("Monday"); break;
            case 2: System.out.println("Tuesday"); break;
            case 3: System.out.println("Wednesday"); break;
            case 4: System.out.println("Thursday"); break;
            case 5: System.out.println("Friday"); break;
            case 6: System.out.println("Saturday"); break;
            case 7: System.out.println("Sunday"); break;
            default: System.out.println("Invalid day number!");
        }
    }
}

```

// Loops

```

    public static void printTable(int num) {
        System.out.println("Multiplication Table of " + num);
        for (int i = 1; i <= 10; i++) {
            System.out.println(num + " x " + i + " = " + (num * i));
        }
    }
}

```

```
public static void displayNumbers(int N) {  
    for (int i = 1; i <= N; i++) System.out.print(i + " ");  
    System.out.println();  
}
```

```
public static void sumNaturalNumbers(int N) {  
    int sum = 0;  
    for (int i = 1; i <= N; i++) sum += i;  
    System.out.println("Sum of first " + N + " natural numbers = " + sum);  
}
```

```
public static void fibonacciSeries(int N) {  
    int a = 0, b = 1;  
    System.out.print("Fibonacci Series: ");  
    for (int i = 1; i <= N; i++) {  
        System.out.print(a + " ");  
        int next = a + b;  
        a = b;  
        b = next;  
    }  
    System.out.println();  
}
```

// methods: with parameters and return value

```
public static int factorial(int num) {  
    int fact = 1;  
    for (int i = 1; i <= num; i++) fact *= i;  
    return fact;  
}
```

```
public static boolean isPrime(int num) {  
    if (num <= 1) return false;  
    for (int i = 2; i <= Math.sqrt(num); i++) {  
        if (num % i == 0) return false;  
    }  
    return true;  
}
```

```
public static int maxOfTwo(int a, int b) {  
    return (a > b) ? a : b;  
}
```

```
public static double areaOfCircle(double radius) {  
    return Math.PI * radius * radius;  
}
```

```
// call each method from main
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    int choice;  
  
    do {  
        System.out.println("\n--- Student Utility Program ---");  
        System.out.println("1. Student Info");  
        System.out.println("2. Even/Odd Check");  
        System.out.println("3. Largest of Three Numbers");  
        System.out.println("4. Display Grade");
```

```
System.out.println("5. Day of Week");
System.out.println("6. Multiplication Table");
System.out.println("7. Display Numbers 1 to N");
System.out.println("8. Sum of Natural Numbers");
System.out.println("9. Fibonacci Series");
System.out.println("10. Factorial");
System.out.println("11. Prime Check");
System.out.println("12. Max of Two Numbers");
System.out.println("13. Area of Circle");
System.out.println("0. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
```

```
switch (choice) {
case 1:
    sc.nextLine(); // consume newline
    System.out.print("Enter Name: ");
    String name = sc.nextLine();
    System.out.print("Enter Roll Number: ");
    int roll = sc.nextInt();
    int[] marks = new int[3];
    System.out.println("Enter 3 subject marks:");
    for (int i = 0; i < 3; i++) marks[i] = sc.nextInt();
    displayInfo(name, roll, marks);
    break;
```

```
case 2:
```

```
System.out.print("Enter number: ");  
evenOdd(sc.nextInt());  
break;
```

case 3:

```
System.out.print("Enter three numbers: ");  
findLargest(sc.nextInt(), sc.nextInt(), sc.nextInt());  
break;
```

case 4:

```
System.out.print("Enter percentage: ");  
displayResult(sc.nextDouble());  
break;
```

case 5:

```
System.out.print("Enter day number (1-7): ");  
dayOfWeek(sc.nextInt());  
break;
```

case 6:

```
System.out.print("Enter number: ");  
printTable(sc.nextInt());  
break;
```

case 7:

```
System.out.print("Enter N: ");  
displayNumbers(sc.nextInt());  
break;
```


case 8:

```
System.out.print("Enter N: ");  
sumNaturalNumbers(sc.nextInt());  
break;
```

case 9:

```
System.out.print("Enter N: ");  
fibonacciSeries(sc.nextInt());  
break;
```

case 10:

```
System.out.print("Enter number: ");  
int num = sc.nextInt();  
System.out.println("Factorial = " + factorial(num));  
break;
```

case 11:

```
System.out.print("Enter number: ");  
int primeNum = sc.nextInt();  
System.out.println(primeNum + (isPrime(primeNum) ? " is Prime" : " is Not  
Prime"));  
break;
```

case 12:

```
System.out.print("Enter two numbers: ");  
System.out.println("Max = " + maxOfTwo(sc.nextInt(), sc.nextInt()));  
break;
```

case 13:

```
System.out.print("Enter radius: ");
```

```
System.out.println("Area = " + areaOfCircle(sc.nextDouble()));
```

```
break;
```

case 0:

```
System.out.println("Exiting program...");
```

```
break;
```

default:

```
System.out.println("Invalid choice!");
```

```
}
```

```
} while (choice != 0);
```

```
sc.close();
```

```
}}
```

```
//observation questions answers
```

```
// question 1
```

```
//Package declaration → package studentutility;
```

```
//Import statements → import java.util.Scanner;
```

```
//Class definition → public class StudentUtilityProgram
```

```
// main method → public static void main(String[] args)
```

```
//Methods → User-defined methods for tasks (factorial, prime check, etc.)
```

```
// So the structure is: Package → Imports → Class → Main → Methods
```

```
// question 2
```

//primitive: Built-in, basic types

// Examples: int, char, double, boolean Examples: String, Array, Class, Object

//Stored directly in memory

//Always lowercase keywords

//Non-Primitive Data Types:Created by programmer or Java library

//Examples: String, Array, Class, Object

// question 3

//if-else

//Used for complex conditions.

//Works with relational operators (<, >, ==, etc.).

//Flexible but can become lengthy.

//switch

//Used when a variable is compared against fixed values.

//Works with int, char, String, enums.

//Cleaner for multiple discrete choices

// question 4

//for loop Used when number of iterations is known.

//while loop Used when condition must be checked before each iteration

//do-while loop Executes at least once, condition checked after iteration

//

Output Screenshot (All Cases)

--- Student Utility Program ---

1. Student Info
2. Even/Odd Check
3. Largest of Three Numbers
4. Display Grade
5. Day of Week
6. Multiplication Table
7. Display Numbers 1 to N
8. Sum of Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Check
12. Max of Two Numbers
13. Area of Circle
0. Exit

Enter your choice: 1

Enter Name: Haarika

Enter Roll Number: 101

Enter 3 subject marks:

80 90 85

Student information :

Name of the student: Haarika

Roll number: 101

Total marks: 255

Percentage: 85.0

Grade: A

Enter your choice: 2

Enter number: 17

17 is Odd

Enter your choice: 3

Enter three numbers: 45 67 23

Largest number is: 67

Enter your choice: 4

Enter percentage: 92

Grade: A+

Enter your choice: 5

Enter day number (1-7): 6

Saturday

Enter your choice: 6

Enter number: 5

Multiplication Table of 5

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

Enter your choice: 7

Enter N: 10

1 2 3 4 5 6 7 8 9 10

Enter your choice: 8

Enter N: 5

Sum of first 5 natural numbers = 15

Enter your choice: 9

Enter N: 7

Fibonacci Series: 0 1 1 2 3 5 8

Enter your choice: 10

Enter number: 6

Factorial = 720

Enter your choice: 11

Enter number: 29

29 is Prime

Enter your choice: 12

Enter two numbers: 45 78

Max = 78

Enter your choice: 13

Enter radius: 7

Area = 153.93804002589985

Enter your choice: 0

Exiting program...